

A Packing of 14 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 14 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.584278. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.571140, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 14 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.584278 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 14$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.571140, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 14/13 = 2.584615$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N14.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
258427826867866548923457475573369423981235656270961905085394490527520405
244112840213795678292501628537275947071018957631804034523184877277186155
8674634090251020430436278395414239834890648537907
```

denominator:

```
999999999997170643860578358508463708313090554265188276773383138716901523
426934839453020109025813286828989970933488113289677548357085065628642577
212813674347159887610477851433256914701622062271
```

B Coordinates

14 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N14.txt`.

```
-324189523068 -553499030535 527568147050 227276075253615524081 345400947164 -320339802628 -1197083183566600581089 489028524610 85207042412 132008065079
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41141849562 -270442148755 -98013557568 -223691674433554097827688 -445680300496 -169852637478 -271160660827140206 410993963605 136670621568 191793363509
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